

CellNet : Counting Cells using Sparse and Noisy Point Annotations

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AIMS AND CHALLENGE

Goal: Understand gene functions and roles in diseases

- ▶ Using Saturation Genome Editing screening (CRISPR based DNA phenotyping)
- ▶ To map the effect of all human genome variants (project MAVE-SGE)

Challenge: Massive Scale

- ▶ ~20,000 genes in human genome
- ▶ × 4 screens per gene (× 3 technical replicate)

▶ **≈ 240,000 screens to perform**

ESTABLISHED PROCESS

Counting cells using dedicated devices

- ▶ Many manual steps
Add trypsin, wait 5mins, add media, resuspend, pipette Trypan blue, load counting slide, count using Countess 3
- ▶ High cost per screen
≥ £30 consumables (+ device cost)
≥ 25mins human labour

COMPUTER VISION BASED COUNTING

Counting cells using microscopy images

- ▶ No additional hardware
- ▶ No single-use consumables
- ▶ Fully automatable

Challenges:

- ▶ Annotation effort (0-shot, masks, bboxes, **points**)
- ▶ Image quality (⇒ **regression based counting**)

THE CELLNET SOLUTION

Fig. 1 Regressing cell probability density

- ▶ Point annotations (1-hot x/y)
- ▶ Spread with Gaussian
- ▶ Mask sparsely annotated regions
- ▶ **Count = integrate over image area**
- ▶ Minimal effort
- ▶ Robust to errors
- ▶ Robust to blurry images

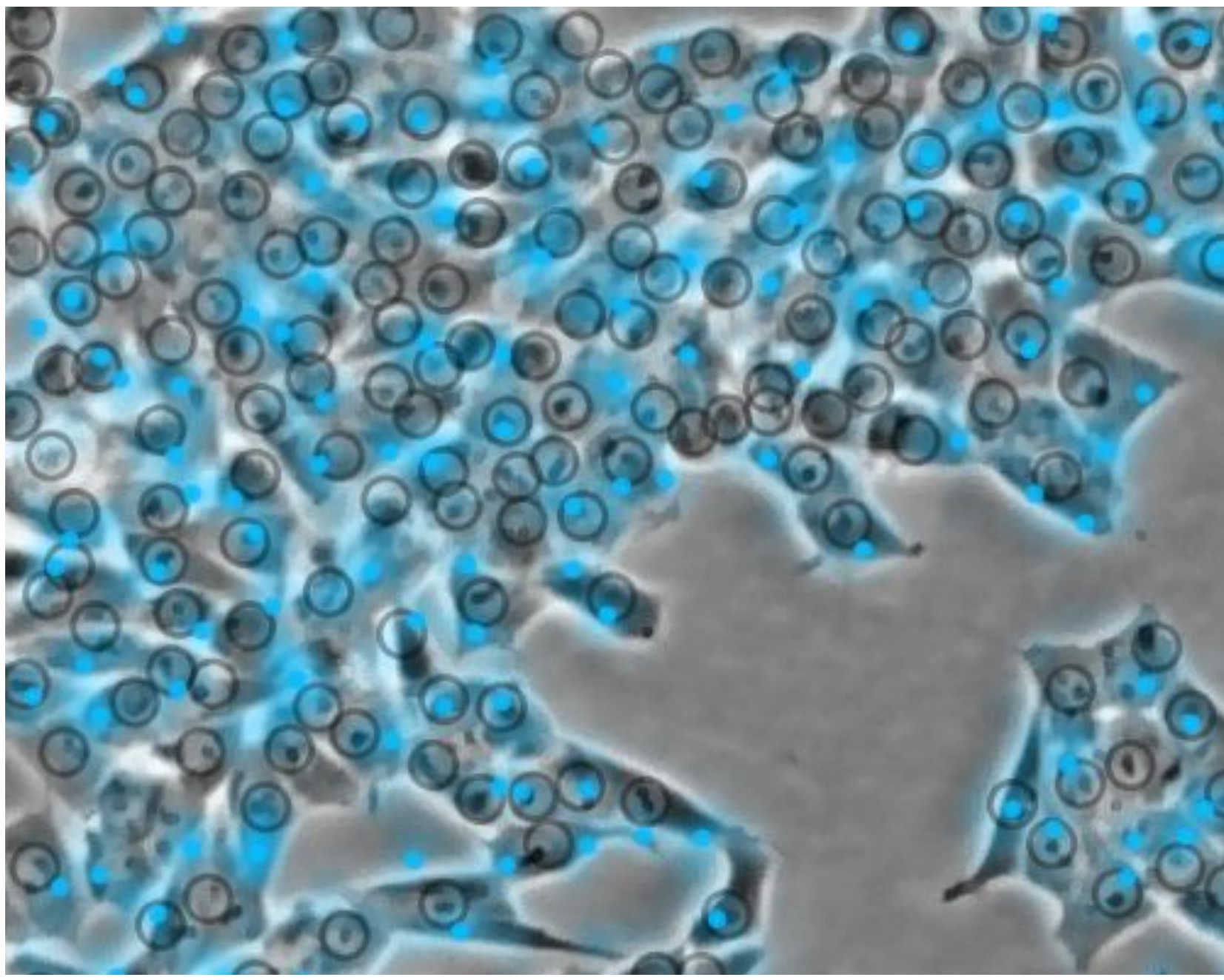


Fig. 2 Image (grey), annotations (circles), prediction (blue), maxima (dots)

```
for x,y in points:
    one_hot_array[x,y] = 1

target = gaussian_filter(one_hot_array, sigma)

augmentations = [
    Crop(),
    Rotate(),
    Mirror(),
    BrightnessContrast(),
    Normalize(),
    ...
]

UNet = UnetPlusPlus(
    classes=1, activation='sigmoid',
    encoder='timm-mobilenetv3_large_100',
    depth=5, channels=(256, 128, 64, 32, 16),
).to('cuda')

for e in epochs:
    for image, target, mask in dataloader:
        prediction = UNet(image)
        loss = MSE(target*mask, prediction*mask)
        ...

counts = prediction.sum()
```

Fig. 3 Pseudo code of the core algorithm

RESULTS

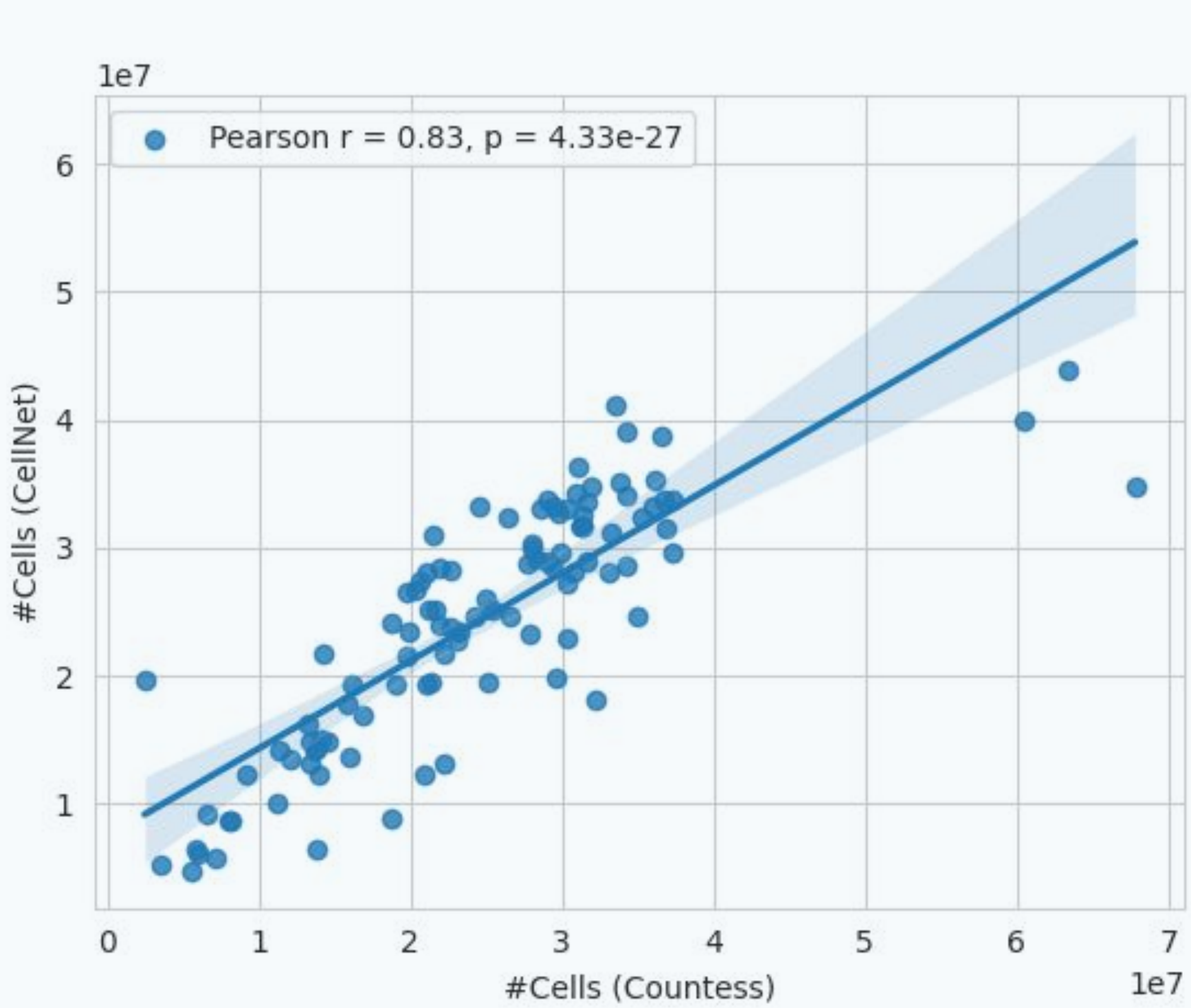


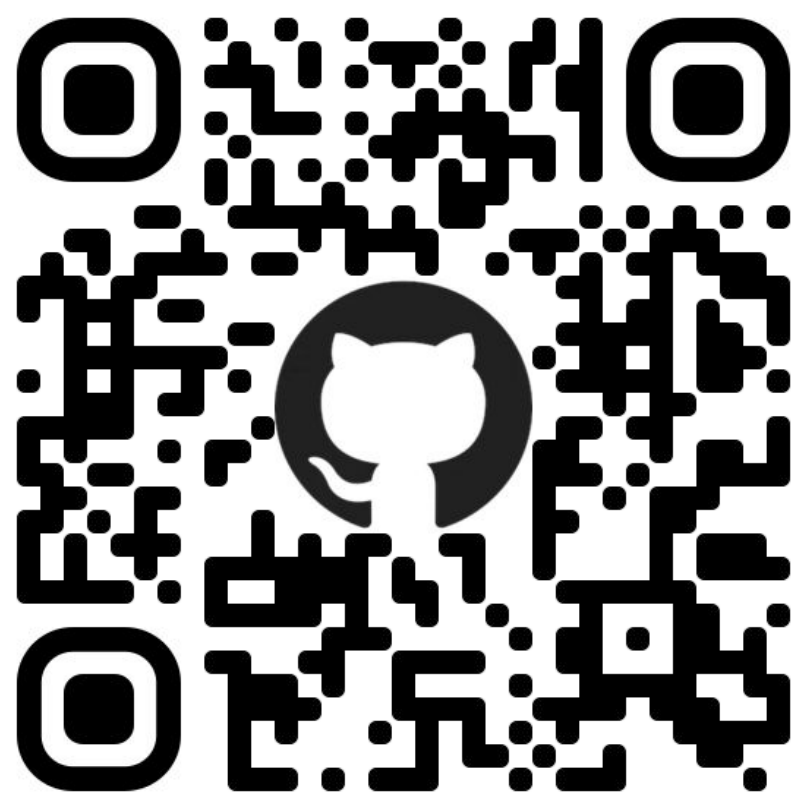
Fig. 4 Correlation between *CellNet* and dedicated device ThermoFisher Countess 3 (n = 105)

Dataset	Model	sMAPE	MAE
LiveCell	CellposeSAM	7.94%	57.80
	CellNet	30.50%	148.06
Sanger23	CellposeSAM	19.09%	452.00
	CellNet	51.04%	1186.12

Fig. 5 Benchmarking *CellNet* against CellposeSAM cell segmentation model LiveCell dataset and own dataset *Sanger23* (n=4)

- ▶ Comparably accurate but much cheaper and faster than dedicated counting device
- ▶ Still a gap to close to other segmentation methods
- ▶ Promising with low quality / low quantity data

Resources



Our source code.



Paper and references.